

Theory of Automata

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Assignment - 1

Date: 10-03-2024

Sun Mon ☒ Tue Wed Thu Fri Sat

$$\Sigma = \{a, b\}$$

1. $aaa, baaa, aaab, baab, aaaaaa, aaaaaaaa, \dots$

$$RE = (b^*aaa)^*b^*$$

2. $(a+b)^*(s_1 + s_2 + s_3 + s_4)(a+b)^*$

$$\Sigma^*(s_1 + s_2 + s_3 + s_4)\Sigma^*$$

3a. exactly 3 b's in total

$$RE = a^*ba^*ba^*ba^*$$

3b. exactly 2 b's or exactly 3 b's, not more

$$RE = a^*ba^*ba^* + a^*ba^*ba^*ba^*$$

4. $\Sigma = \{a, b\}$

a. All strings that end in a double letter.

$$RE = (a+b)^*(aa+bb)$$

b. All strings that have exactly one double letter in them.

$$RE = (a+b)^*(aa+bb)(a+b)^*$$

5. All strings in which letter b is never tripled. No word contains the substring bbb.

$$RE = (a+ba+bba)^*$$

6. All words in which a is tripled or b is tripled, but not both.

$$RE = (a+b)^*aaa(a+b)^* + (a+b)^*bbb(a+b)^*$$

7. $\Sigma = \{a, b\}$

a. All strings that do not have the substring ab.

$$RE = b^*a^*$$

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7b. All strings that do not have both the substrings bba and abb.

L_1 : strings without bba $\Rightarrow (a+b)^* - (a+b)^* bba (a+b)^*$

L_2 : strings without abb $\Rightarrow (a+b)^* - (a+b)^* abb (a+b)^*$

$$\Rightarrow L = L_1 \cup L_2 \quad RE = [(a+b)^* - (a+b)^* bba (a+b)^*] + [(a+b)^* - (a+b)^* abb (a+b)^*]$$

OR

$$RE = (a+ba)^* b^* \cup b^* a (a+ba)^* (b|\lambda)$$

8. All strings in which the total number of a's is divisible by three, such as aabaabbaba.

$$RE = (b^* aaa)^* b^*$$

9. $\Sigma = \{a, b\}$

a. All strings in which any b's that occur are found in clumps of ~~an~~ odd number at a time.

$$RE = (a^* b (bb)^* a^*)^*$$

b. All strings that have an even number of a's and an odd number of b's.

$$\text{even a's} = (aa)^* \quad \text{odd b's} = b(bb)^*$$

$$RE = (aa)^* b (bb)^* (aa)^*$$

c. All strings that have an odd number of a's and an odd number of b's.

$$\text{odd a's} = a(aa)^* \quad \text{odd b's} = b(bb)^*$$

$$RE = a(aa)^* b(bb)^*$$

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10. $(a+b)^* a(a+b)^* b(a+b)^* \Rightarrow ab, aab, abb, abab, baab, aaabb, aba, \dots$

a. Show that this is equivalent to

$(a+b)^* ab(a+b)^*$ in the sense that they define the same language.

$\hookrightarrow ab, aab, abb, abab, baab, aaabb, \dots$

Both languages have strings containing substring "ab"

$$\therefore (a+b)^* a(a+b)^* b(a+b)^* = (a+b)^* ab(a+b)^*$$

b. Show that $(a+b)^* ab(a+b)^* + b^* a^* = (a+b)^*$

$$RE_1 \rightarrow (a+b)^* ab(a+b)^* + b^* a^*$$

• part 1: $(a+b)^* ab(a+b)^*$ contains 'ab'

$L = ab, aab, abb, abab, baab, \dots$

• part 2: $b^* a^*$ does not contain 'ab'

$L = \Lambda, b, bb, a, aa, ba, bba, bbaa, \dots$

Language of RE_1 every possible string over a, b

$$L_1 = (a+b)^*$$

$$RE_2 \rightarrow (a+b)^*$$

language of RE_2 every possible string over a, b

$$L_2 = (a+b)^*$$

$$L_1 = L_2 = (a+b)^*$$

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10.

c. Show that

$$(a+b)^* ab[(a+b)^* ab(a+b)^* + b^* a^*] + b^* a^* = (a+b)^*$$

Already proved in part b

Final Expression : $(a+b)^*$

all possible strings over a,b

$L \Rightarrow \Lambda, a, b, ab, ba, aa, bb, aba, bab, aabb, \dots$

11. Check RE equivalence:

a. $(ab)^* a$ and $a(ba)^*$ ✓

• $\rightarrow a, aba, ababa, abababa, \dots$

• $\rightarrow a, aba, ababa, abababa, \dots$

b. $(a^* + b)^*$ and $(a+b)^*$ ✓

• $a^* \rightarrow$ any number of a's, and with a $\rightarrow$ repeat all strings over a,b

• $(a+b)^* \rightarrow$ all strings over a,b

c. $(a^* + b^*)^*$ and $(a+b)^*$ ✓

• $a^* \rightarrow$ all a's, $b^* \rightarrow$ all b's $\rightarrow$ repeat all possible strings over a,b

• $(a+b)^* \rightarrow$ all strings over a,b

d. ϵ^* and ϵ ✓

• $\epsilon^* = \epsilon$

e. $(a^* b)^* a^*$ and $a^* (b a^*)^*$ ✓

• Both $\rightarrow$ all strings over a,b with any number of a's

• Language $\rightarrow \Lambda, a, aa, b, ab, aab, \dots$

f. $(a^* bbb)^* a^*$ and $a^* (bbba^*)^*$ ✓

• Both $\rightarrow$ b's appear only in group of 3, any number of a's

• Language $\rightarrow \Lambda, a, aaa, bbb, aaabbb, bbbaaa, \dots$

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11.

g. $((a+bb)^*aa)^*$ and $\epsilon + (a+bb)^*aa$ X

$RE_1 \rightarrow \Lambda, a, aaaa, abaa, aabaa, bbaaaa, \dots$ whole can repeat

$RE_2 \rightarrow \epsilon$ or one string ends in aa

$\Lambda, aa, bbaa, aabaa, abbaa, \dots$ only one block, not repeated

h. $(aa)^*(\epsilon + a)$ and a^* ✓

• RE_1 = even number of a's and option one extra a

• $RE_2 = a, aa, aaaa, aaaa, \dots$

i. $a(aa)^*(\epsilon + a)b + b$ and a^*b X

• First = $b, ab, aab, abab, ababb, \dots$

• Second = $b, ab, aab, aab, aaaaab, \dots$

j. $a(ba + a)^*b$ and $aa^*b(aa^*b)^*$ ✓

• First = $ab, aab, aaab, aaaaab, abab, aabab, \dots$

• Second = $ab, aab, aab, aab, aab, aabab, \dots$

k. $\epsilon + a(a+b)^* + (a+b)^*aa(a+b)^*$ and $((b^*a)^*ab^*)^*$ X

• First : $\Lambda, a, ab, aba, abb, aa, baa, bbaa, babaa$

• Second : $\Lambda, a, ab, ba, aba, bab, aa, baa, baba$

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12.

a. $(a+b)^* a (\varepsilon + bbbb)$

English: All strings over alphabet that end with a bbbb

Language: a bbbb, aabbbb, baaaa,

b. $(a(a+bb)^*)^*$

All strings made by repeating a or bbb any number of times

Λ, a, bbb, aa, abbb, bbaa,

c. $(a(aa)^* b(bb)^*)^*$

All strings having any number of aaaa blocks followed by any number of bbb.

Λ, aaa, bbb, aabbb, aaaaaabbbbb,

d. $(b(bb)^*)^* (a(aa)^* b(bb)^*)^*$

Start with one b, have any number of bbb, one a, any number of aa

ba, baa, baaa, bbbba,

e. $(b(bb)^*)^* (a(aa)^* b(bb)^*)^* (a(aa)^*)^*$

Start with b, have groups of bbb, one a, groups of aa, one b, groups of bb, groups of aaaa

bab, babaa, bbbbaab,

f. $((a+ba)a)^*$

All strings made by repeating 2-letter block whose 2nd letter is always a.

Λ, aa, ba, aaba, baaa, baba,